Fast Human–Robot Object Handover with Stereo Vision and Tactile Closed-Loop Control Technical Report

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***Abstract*—This report presents a fast human–robot object handover system. The system uses iceoryx2 instead of ROS as its single-host, multi-process communication layer and acquires color, left/right infrared, and depth streams from two RealSense D455 cameras. Fast-FoundationStereo depth estimation, WiLoR hand-model estimation, dual-camera palm-position fusion, tabletop object point-cloud segmentation, control of a seven-degree-of-freedom Nero arm, and PaXini 3-D tactile feedback are integrated to realize a rapid handover process from human contact with an unknown object to robot grasping, transport, and placement. The control architecture consists of a visual outer loop, a manipulator inner loop, and an independent tactile loop. The visual outer loop generates the desired end-effector target; the manipulator inner loop performs reachability constraints, inverse kinematics, and joint-rate-limited tracking; and the tactile loop handles gripper contact, grasp confirmation, slip monitoring, and compliant release. Completing one handover within five seconds after triggering is retained as a task-level design objective.**

I. System Overview

The system is designed for fast and stable human–robot object handover in tabletop scenes. Its inputs include color, left/right infrared, and depth streams from two RealSense D455 cameras, together with 3-D force signals from PaXini tactile sensors mounted on both gripper fingers. Its outputs include the desired Nero end-effector target, joint commands, and gripper opening/closing commands. iceoryx2 provides inter-process communication so that perception, tactile sensing, the task state machine, and manipulator servoing can run as decoupled processes.

Fig. 1 presents the complete system. The dual-camera vision processes continuously publish hand poses and object point clouds; the tactile process publishes force feedback from both fingertips; the task state machine selects the waiting, tracking, grasping, and placement phases according to the relationship between the hand and object; and the Nero servo process executes both the manipulator inner loop and the tactile gripper loop. This architecture avoids single-process blocking and allows the perception, tactile, and control modules to be debugged independently.

The Nero base frame $\mathcal{F}_B$ is used as the common world reference frame. Each D455 depth map is first back-projected into a point cloud in its camera frame $\mathcal{F}_{C_i}$ and then transformed to $\mathcal{F}_B$ using the calibrated extrinsic matrix ${}^{B}\mathbf{T}_{C_i}$. Hand keypoints and the palm position are transformed through the same extrinsic chain. Hereafter, the terms “manipulator base frame,” “Nero base frame,” and “world frame” all refer to $\mathcal{F}_B$, which is used for hand–object contact tests, recording the initial object position, and generating manipulator targets.

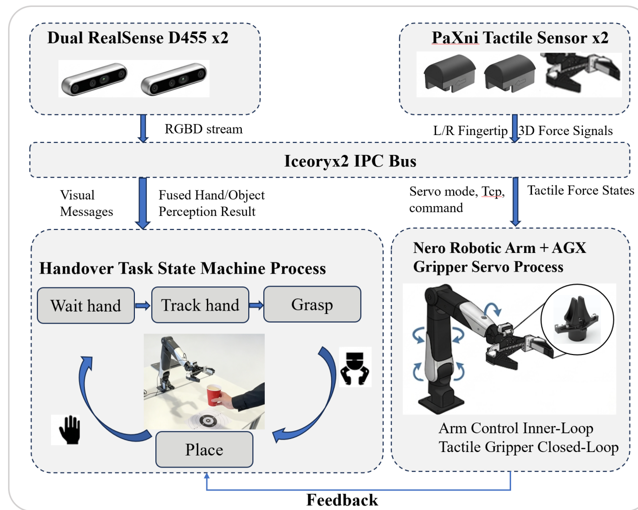

Fig. 1. System overview. Two D455 cameras, PaXini tactile sensing, the iceoryx2 bus, the task state machine, and the Nero servo jointly form the closed-loop handover system.

II. Problem Analysis

The central difficulty of unknown-object handover is that the system cannot rely on a pre-modeled object category, shape, or grasp point. The robot observes a mixture of the human hand, tabletop, unknown object, and occluded regions. On the vision side, two information-quality problems dominate. First, the hand pose and hand-model quality determine the reliability of contact triggering, palm motion direction, the target region in front of the hand, and hand-point removal. Second, depth-map and point-cloud quality determine the stability of the object-cloud centroid, table segmentation, and placement height. Hand-estimation jitter causes the manipulator to track an incorrect target, while missing depth on transparent, reflective, thin, or occluded objects fragments the object cloud and degrades both grasp-point and placement-height estimates.

The system does not depend on category recognition, a prebuilt object model, or a predefined grasp point. Instead, it uses the hand–object contact constraint to generate a local geometric target. A hand-contact event triggers the task, the hand model removes points occupied by the hand, the depth cloud estimates local object geometry, and the manipulator then tracks a pre-grasp region offset from the object-cloud centroid. WiLoR provides in-the-wild 3-D hand localization and reconstruction [1]. The FoundationStereo family provides strong zero-shot stereo depth estimation, while FastFoundationStereo is used as the engineering depth backend for infrared stereo depth completion and improved aligned-depth quality [2].

III. Visual Outer Loop

The visual outer loop generates the hand position and object target from the dual-camera observations. Each D455 simultaneously acquires color, left/right infrared, and depth data. FastFoundationStereo processes the two infrared images to produce a depth map aligned with the color image, while WiLoR processes the color image to estimate 21 hand keypoints and the complete hand model. Within each camera, five palm keypoints are projected into the depth map to recover metric depth and are then fused with the 3-D keypoints predicted by WiLoR. Let the palm index set be $\mathcal{I} = \{0, 5, 9, 13, 17\}$, the pixel coordinate be (u_i, v_i), the depth be z_i, and the camera intrinsic matrix be K. Then

$$\begin{aligned} \mathbf{p}_i^d &= z_i K^{-1} \begin{bmatrix} u_i & v_i & 1 \end{bmatrix}^T, \\ \mathbf{p}_i &= \lambda \mathbf{p}_i^d + (1-\lambda)\mathbf{p}_i^w, \quad \mathbf{c}_h = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbf{p}_i. \end{aligned} \tag{1}$$

Here, $\mathbf{p}_i^d$ is the depth-back-projected 3-D point, $\mathbf{p}_i^w$ is the WiLoR 3-D keypoint, λ is the depth-branch weight, and $\mathbf{c}_h$ is the palm center.

After each camera produces a 3-D palm position, a constant-velocity Kalman model fuses the two position observations. The velocity term is projected forward to compensate for acquisition, inference, and communication delay. The state is $\mathbf{x}_k = [\mathbf{p}_k^T, \mathbf{v}_k^T]^T$, the measurement $\mathbf{z}_k$ is the palm position reported by a camera, and the prediction gain γ corresponds to the configuration item `hand_fusion_prediction_gain`:

$$\begin{aligned} \mathbf{x}_k^- &= \begin{bmatrix} I & \gamma I \\ 0 & I \end{bmatrix} \mathbf{x}_{k-1}, \\ \mathbf{z}_k &= \begin{bmatrix} I & 0 \end{bmatrix} \mathbf{x}_k + \mathbf{n}_k, \quad \hat{\mathbf{p}}_k = \mathbf{p}_k + \tau_s \mathbf{v}_k. \end{aligned} \tag{2}$$

The predicted palm position $\hat{\mathbf{p}}_k$ compensates for the total system delay τ_s before publication to downstream modules. Compared with a single-camera observation, dual-camera fusion reduces jitter caused by occlusion, local depth loss, and single-view errors. Velocity prediction also places the manipulator target closer to the current hand motion rather than to a delayed model output. Fig. 2 summarizes the visual outer loop.

The object target is obtained from the depth cloud. Depth maps from the two cameras are back-projected into point clouds, and only points inside the fixed tabletop workspace are retained. Table-plane segmentation extracts points above the table, and the WiLoR hand-model cloud dynamically removes the region occupied by the hand. The remaining points are used to estimate the local point-cloud centroid. Rather than tracking the human hand directly, the manipulator tracks a point located ahead of the centroid of the object cloud being contacted and pushed by the hand. Only the 3-D target position is updated.

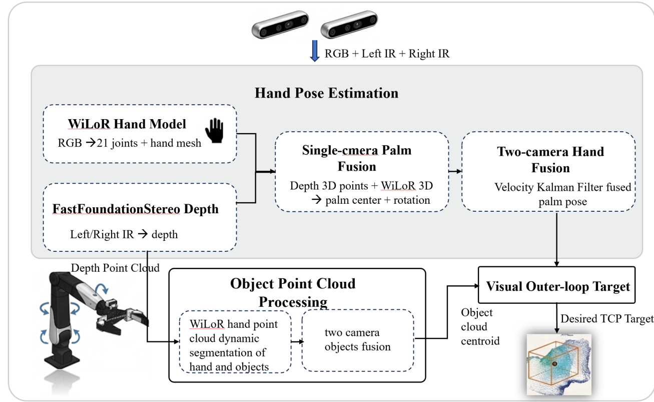

Fig. 2. Visual outer loop. The two D455 observations pass through the FastFoundationStereo depth branch, WiLoR hand-model branch, palm 3-D fusion, dual-camera Kalman fusion, and object point-cloud segmentation to generate the visual target.

IV. Hand–Object Trigger and Manipulator Hardware

The task is triggered by the contact relationship between the hand-model cloud and the tabletop object cloud. During the waiting phase, the manipulator remains at its initial pose, the gripper stays open, and the vision system continuously updates the tabletop object cloud. When contact between the two clouds is detected, the system records the current object position and transitions from waiting to tracking. This design allows the placement phase to return near the object's initial region while avoiding repeated use of an unstable initial estimate after the object starts moving.

The object cloud is obtained by fusing the left and right depth clouds, cropping them to the workspace, removing the table plane, and removing the hand-model cloud:

$$\begin{aligned} \mathcal{P}_o &= \mathrm{Crop}(\mathcal{P}_L \cup \mathcal{P}_R) \setminus (\mathcal{P}_{\text{table}} \cup \mathcal{P}_{\text{hand}}), \\ \mathbf{c}_o &= \frac{1}{|\mathcal{P}_o|} \sum_{\mathbf{p} \in \mathcal{P}_o} \mathbf{p}. \end{aligned} \tag{3}$$

Here, $\mathcal{P}_o$ is the candidate object cloud after table and hand removal, and $\mathbf{c}_o$ is its arithmetic centroid. The pre-grasp target is $\mathbf{x}^* = (\mathbf{p}_g^*, \mathbf{R}_g)$, where $\mathbf{p}_g^* = \mathbf{c}_o + \mathbf{d}_g$. The offset $\mathbf{d}_g$ follows either the human-hand motion direction or the gripper approach direction. The end-effector rotation matrix $\mathbf{R}_g$ is predefined and remains fixed throughout tracking. Consequently, the visual outer loop updates only $\mathbf{p}_g^*$ and does not update the target orientation.

Fig. 3 shows the manipulator hardware. The seven-degree-of-freedom Nero arm executes the end-effector trajectory, the AGX gripper performs grasping and release, and PaXini tactile modules mounted on both fingers provide closed-loop grasp feedback. The dual cameras provide external perception, the manipulator performs the motion, and the tactile modules report contact states.

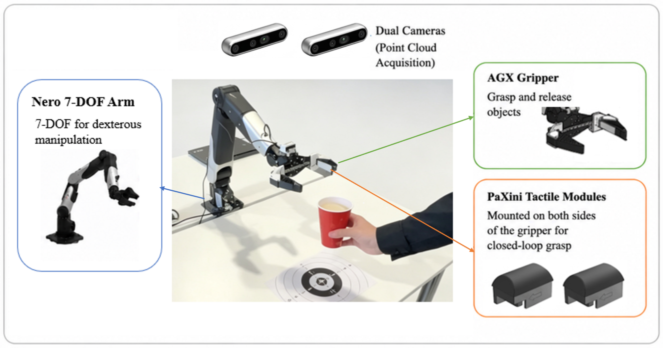

Fig. 3. Manipulator hardware. The seven-degree-of-freedom Nero arm, AGX gripper, and PaXini tactile modules jointly execute grasping, transport, and release.

V. Three-Loop Control Architecture

The control system contains three loops, as shown in Fig. 4. The visual outer loop generates the desired end-effector target from the fused palm position and object cloud. The manipulator inner loop converts this target into stable Nero joint commands through workspace bounds, reachability projection, inverse kinematics, joint-step limiting, and stale-target holding. The independent tactile loop controls gripper opening and closing from the PaXini 3-D force signals and detects contact, grasp success, slip tendency, and table contact.

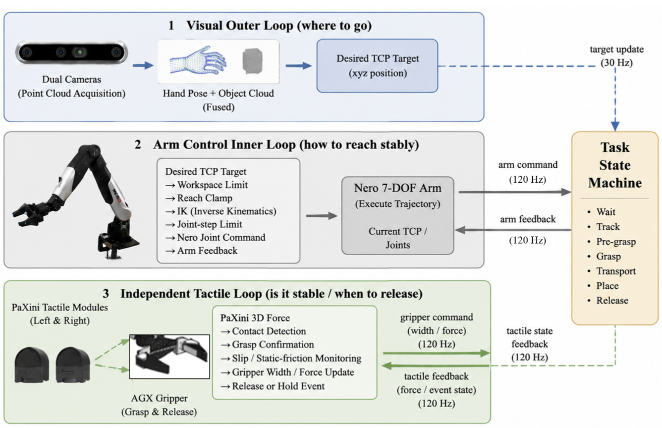

Fig. 4. Three-loop control architecture. The visual outer loop, manipulator inner loop, and independent tactile loop are responsible for target generation, stable tracking, and contact control, respectively.

The visual outer loop determines where to move. During tracking, it generates the pre-grasp target from the object-cloud centroid and the region in front of the hand. When live height updates are required, the planar position can follow the fused hand position while the height is estimated from the local cloud in front of the hand. The manipulator inner loop determines how to reach the target stably. It follows a real-time visual-servo design analogous to accepting end-effector pose or velocity targets in MoveIt Servo: every servo cycle reads the latest end-effector target, checks whether it is stale, solves inverse kinematics, and limits the per-cycle joint step to prevent abrupt motion caused by target jitter. The inner-loop update is

$$\begin{aligned}
\mathbf{e}_k &= \mathrm{PoseError}(\mathbf{x}_k^*, f(\mathbf{q}_k)), \\
\Delta\mathbf{q}_k &= \mathrm{clip}(J_\lambda^\dagger \mathbf{e}_k, \pm\Delta\mathbf{q}_{\max}), \\
\mathbf{q}_{k+1} &= \mathbf{q}_k + \Delta\mathbf{q}_k.
\end{aligned} \tag{4}$$

The operator $\mathrm{PoseError}(\cdot,\cdot)$ maps the target and current end-effector poses to a Cartesian error. Because $\mathbf{R}_g$ is fixed, visual updates change only the position component of this error. $J_\lambda^\dagger$ is the damped pseudoinverse, $f(\cdot)$ is the forward-kinematics function, and $\Delta\mathbf{q}_{\max}$ is the maximum joint increment permitted in one control cycle at the 90 Hz servo rate. The tactile loop determines whether the object is stably grasped and when it should be released. It does not directly command the end-effector position; instead, gripper width, grip force, and tactile events influence the grasp and placement states.

The tactile loop further applies impedance-based gripper adjustment. It reads the normal force, tangential-force ratio, downward-slip ratio, and force derivative from the two PaXini sensors. As downward-slip risk, tangential risk, or insufficient normal support increases, a virtual-spring relation reduces the desired gripper width and adds a dynamic compression term. The empirically tuned parameters used here are the tangential-risk target $\rho^* = 0.35$, downward-slip target $r_d^* = 0.30$, normal-force error gain $k_n = 0.003\,\mathrm{m/N}$, and maximum dynamic compression $d_{\max} = 0.0015\,\mathrm{m}$. This loop compensates for insufficient static friction during transport and releases the object early when table contact is detected during placement.

$$\begin{aligned}
s_k &= \max\left(\frac{\rho}{\rho^*}, \frac{r_d}{r_d^*}\right), \\
\delta w_k &= k_n[n^* - n]_+ + d_{\max}\tanh([s_k - 1]_+), \\
w_{k+1} &= \mathrm{clip}(w_{\mathrm{ref}} - \delta w_k, w_{\min}, w_{\max}).
\end{aligned} \tag{5}$$

Here, ρ and r_d denote tangential and downward-slip risks, respectively, and δw_k is the additional gripper compression produced by the virtual spring. As risk increases, the target width decreases to improve grasp stability. When risk disappears, the width returns to its reference value to avoid excessive compression of an unknown object.

VI. Timing and Communication Design

The system requires the effective execution time from hand–object contact to object release to remain within five seconds. Inference delay, communication delay, and servo period must therefore be controlled jointly. WiLoR and Fast-FoundationStereo are computationally intensive vision models and consume most of the latency budget. Repeated serialization, copying, and scheduling queues for images, depth maps, and point clouds would reduce the time available for manipulator tracking and tactile release. iceoryx2 is a zero-copy, lock-free inter-process communication middleware for publish–subscribe and related patterns. The system therefore uses iceoryx2 to organize the data paths among perception, task, tactile, and servo processes. Compared with passing large images and point clouds directly through a ROS topic chain, this design aims to reduce additional communication overhead beyond inference latency.

TABLE I
Main Control Loops and Publication Rates

Module	Rate	Function
D455 capture and visual fusion	30 Hz	Update hand pose, depth, and object cloud
Object-cloud fusion	30 Hz	Synchronize and fuse two camera clouds
Task-target publication	30 Hz	Publish end-effector targets
Nero servo inner loop	90 Hz	Inverse kinematics, limiting, and joint commands
PaXini tactile sampling	120 Hz	Observe 3-D forces at both fingertips
Tactile impedance loop	120 Hz	Assess slip risk, grip force, and release conditions
Gripper command output	30 Hz	Update width and force targets

VII. Task State Machine

A complete task consists of waiting, tracking, grasping, placement, release, and return to the initial pose, as shown in Fig. 5. During waiting, the manipulator stays at its initial pose, the gripper is open, and the tactile sensors are zeroed. The vision system continuously checks contact between the hand model and the tabletop object cloud. Once contact occurs, the initial object position is recorded and the system enters the tracking phase.

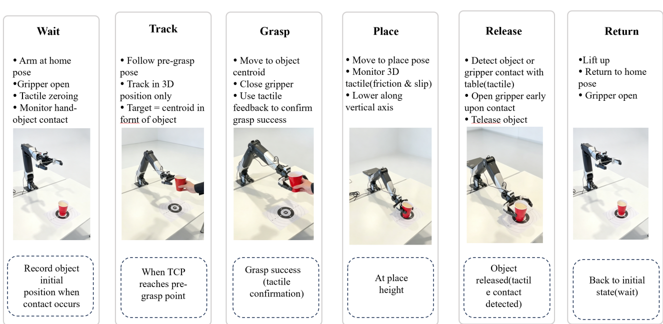

Fig. 5. Task state machine. The system completes one fast handover in the sequence of waiting, tracking, grasping, placement, release, and return to the initial pose.

During tracking, the manipulator follows a point offset from the object-cloud centroid, maintains a fixed pre-grasp orientation, and tracks only the 3-D position. After the end effector reaches the pre-grasp point, the system enters the grasp phase: the manipulator advances toward the object-cloud centroid, the gripper closes, and tactile feedback determines whether the grasp is successful. During placement, the gripper continuously senses 3-D tactile changes and monitors static friction and slip tendency during motion. After the manipulator reaches a position above the placement point, it descends vertically. If tactile sensing detects contact between the object or gripper and the table, the gripper opens early to release the object. The manipulator then lifts and returns to its initial pose.

VIII. Implementation Highlights

The main design contributions are: 1) decoupling the perception, tactile, task, and servo processes with iceoryx2; 2) combining the hand model and stereo depth under a hand–object contact constraint to generate local geometric targets for unknown objects; and 3) constructing a layered control architecture with a visual outer loop, manipulator inner loop, and tactile loop. The loop rates are summarized in Table I.

IX. Conclusion

This report presents the technical design of a fast handover system. Dual-D455 visual sensing, the WiLoR hand model, Fast-FoundationStereo depth estimation, object point-cloud segmentation, Nero visual servoing, and PaXini tactile impedance control are combined for tabletop handover of unknown objects. Completing the task within five seconds remains a design objective, supported by layered control with a 30 Hz visual outer loop, 90 Hz manipulator inner loop, and 120 Hz tactile loop. Future experiments will quantify perception frame rate, target-tracking error, grasp success rate, tactile-detection latency, and total task duration.

References

[1] R. A. Potamias, J. Zhang, J. Deng, and S. Zafeiriou, "WiLoR: End-to-End 3D Hand Localization and Reconstruction in the Wild," arXiv:2409.12259, 2024.

[2] B. Wen, S. Dewan, and S. Birchfield, "Fast-FoundationStereo: Real-Time Zero-Shot Stereo Matching," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2026.